

DASE7506 · MP1 Lab Report

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Abstract

In this experiment, under the fixed WikiText-2 / BPE-2048 protocol, a small autoregressive language model was trained from random initialization. The improved model adopted an 8-layer, 256-dimensional backbone, combined with Partial RoPE, LayerNorm, SwiGLU, learnable replication branch, residual Dropout, Muon/AdamW mixed optimization and EMA. Through nine sets of experiments with equal training target numbers, the model that only applied exclusive self-attention (XSA) in the last 6 layers was selected. Its CPU FP32 verification BPB was 1.462978, and the complete test BPB was 1.477793. The original baseline test BPB was 2.101260. The test BPB decreased by 0.623467 absolutely and 29.67% relatively.

The ablation results support the effectiveness of the replication branch, regularization, EMA, and appropriate XSA under the current configuration. The baseline and the improved model have different scales and training volumes, and the overall benefits cannot be attributed to the causal effect of a single mechanism. This paper also records the search cost, resource measurement gap, and the limitations of a single random seed.

1. Task and Evaluation Agreement

According to code/README.md and GUIDE.md, all models utilize only the training texts and tokenizers provided with the assignment. The validation set is used for configuration and checkpoint selection, while the test set is used for comprehensive evaluation after the freezing procedure. The protocol used is 7506-mp1-wt2-v2, employing an independent causal window of length 256 that includes the final short window, without state transfer across windows.

The evaluation metrics are defined as follows: B denotes the number of original UTF-8 bytes, N denotes the number of scored targets, and NLL is expressed in natural logarithms.

$$\text{BPB} = \frac{\text{NLL}_{\text{nats}}}{B \ln 2}$$
$$\text{PPL}_{\text{token}} = \exp\left(\frac{\text{NLL}_{\text{nats}}}{N}\right)$$

The lower the values, the better. However, for this assignment ranking, BPB is used, and the PPL token cannot be directly compared with the word-level perplexity of publicly available papers.

Data Split	Scoring target number	Original UTF-8 byte count
Validation	376,599	1,148,007
Test	428,405	1,292,013

2. Model Architecture and Design Rationale

The original model had limited capacity, and its standard vocabulary output did not explicitly leverage local text repetitions. In this approach, we simultaneously increase the capacity of the backbone network, introduce causal replication paths, and adjust the informational composition of the attention outputs. These constitute our design hypotheses, the validity of which will be rigorously validated in subsequent comparative experiments.

Component	Final Configuration
Vocabulary / Window	2048 / 256
Body / Attention	width=256, depth=8, heads=4. standard MHA & XSA
Normalization / FFN	LayerNorm. SwiGLU, hidden=688
Positional encoding	Partial RoPE, fraction=0.25, theta=10000
XSA	Standard attention in the first 2 layers, XSA in the subsequent 6 layers
Copy / Regularization	copy_dim=32, Gateable. Dropout=0.1
Weight sharing / Number of parameters	The word embeddings and the output head share the same weights. / 6,895,105 parameters

Partial RoPE: A rotational position encoding is applied to 16 out of the 64 dimensions of each attention head, while the remaining dimensions remain unrotated. The positional rotation is applied exclusively to Q and K. In this experiment, we compare partial-dimensional rotation versus full-dimensional rotation, rather than conducting independent ablation studies between RoPE and absolute position encoding.[3]

The feedforward network of SwiGLU is defined as follows, where g, u, and d represent the gating, dimensionality expansion, and dimensionality reduction projections, respectively.

$$\text{FFN}(h)=W_d[\text{SiLU}(W_g h+b_g)\odot(W_u h+b_u)]+b_d$$

LayerNorm is currently retained, while RMSNorm is not used. Dropout is applied to the attention output projection and the FFN output, respectively, prior to the residual addition.[4]

XSA: First compute the causal attention output, then normalize the value vector of the current token and remove the projection along the corresponding direction.

$$u_t=\frac{v_t}{\max(\|v_t\|_2,\epsilon)}$$

$$a'_t=a_t-(a_t^T u_t)u_t$$

Here, a_t represents the original attention output, v_t is the value vector of the current token, and ϵ is a numerical stability constant. This operation removes the components along the direction of the current token's value. It does not directly mask the diagonal of the attention matrix. In the implementation, this projection is calculated in FP32 format and does not increase the number of learnable parameters.[2]

Copy branch: Using the 32-dimensional Q/K of the final latent state, calculate causal attention on the current and previous observed tokens, and then aggregate them into a word distribution based on token IDs. The gating and mixed distribution are defined as:

$$g_t=\sigma(w^T h_t+b)$$

$$p_t=(1-g_t)p_{\text{vocab},t}+g_t p_{\text{copy},t}$$

$$g_{\text{init}}=0.1$$

Here, σ represents the sigmoid function, and h_t is the final hidden state. The calculation is carried out in the logarithmic domain to maintain stability. It is legal to copy the current input token when predicting the next token.

The structure generates all temporary states in a single forward pass. There is no cross-window retrieval of the library, pre-trained weights, SmearGate, GQA or quantization. The copy branch adds 16,641 parameters. The forward function returns the normalized log probabilities, which can be used as the input for cross-entropy. predict_log_probs uses the same distribution.

3. Experimental setup and comparison design

item	Common configuration of the nine ablation zones
Training Equipment / Software	RTX 4090 D. PyTorch 2.7.1+cu126. FP32
Random seed / CPU thread settings	seed=17. Training threads=16
Update Step / Batch / Window	3000 / 128 / 256
Process the number of training targets	98,304,000 / Gruop
Validation and Point Selection	Every 100 steps. Save and validate BPB for optimal performance raw/EMA
Learning Rate / Scheduling	AdamW peak 0.002. warmup=100. cosine. Minimum ratio 0.1
Muon	lr=0.02. momentum=0.95. NS=5. weight decay=0.01
AdamW / Gradient clipping	weight decay=0.1. group_weight_decay=False. clip=1.0
EMA	Step 400 begins. decay=0.99

Muon handles non-embedded, non-gated 2D matrix parameters. Embedded parameters, shared output weights, gating mechanisms, and other parameters are handled by AdamW. When Muon is disabled, all parameters are handled by AdamW. Both optimizers are scaled proportionally according to the same learning rate. Consequently, this experiment compares the complete optimization formulation, rather than replacing the update rules under exactly the same learning rate and decay parameters.[5]

The "warmdown_steps" value of 700 in the record only takes effect in the linear warmdown scheduling. In this batch of experiments, we use cosine scheduling, and thus we cannot claim that the last 700 steps are executed with linear cooling. EMA is the exponential moving average of the trained weights. During evaluation, only an average weight model is needed, and it does not belong to the multi-model inference integration.

The comparative design consists of three parts:

- ① The original baseline and the end-to-end results of the selected model.
- ② The comparison of the number of XSA layers and the optimizer under the same scale and training target budget.
- ③ The conditional ablation of sequentially removing XSA, copying branches, Partial RoPE, and Dropout. All formal ablations are executed for 3000 steps, and then checkpoints are selected according to the same validation rules.

The original baseline had only 9,830,400 processing targets and 1,088,256 parameters. The number of parameters in the improved model is approximately 6.34 times that of the original. Evidence for the equal budget mechanism should come from comparisons among nine ablation groups, and cannot be directly derived from the gap between the baseline and the final model. Selecting early checkpoints does not offset the computational consumption for the search that follows.

4. Primary outcomes and itemized ablation

Model	Validation BPB	Test BPB	Test token PPL
Raw baseline	2.071083	2.101260	80.8471
Final L6 XSA + EMA	1.462978	1.477793	21.9603

The baseline verification values are derived from the "baseline/metrics.json" file. The improved model verification values are obtained from the independent CPU FP32 scoring file. The test comparison uses the complete test results of both "test_cpu_fp32.json". The final test score is approximately 0.014815 BPB higher than the verification score, but since they come from different data partitions, this cannot be interpreted as a training degradation.

The adjacent difference Δ in the table is "Current row BPB - Previous row BPB", and a positive value indicates a decrease after removing this item. "Full" means that all 8 layers use XSA, not the selected L6.

Systematic design	CPU Validation BPB	Adjacent Δ BPB	Optimal EMA step
Complete Formula (XSA 8)	1.467045	—	1400
Remove XSA	1.476084	+0.009038	1500
Remove XSA & Copy	1.515537	+0.039453	1500
Change to Full RoPE	1.525419	+0.009882	1500
Remove Dropout	1.579223	+0.053805	800

Copy Branch: Without XSA, retaining Partial RoPE and Dropout, enabling copying reduced the BPB from 1.515537 to 1.476084, improving by 0.039453. It provides additional probability paths for local repeated tokens, which is consistent with the motivation. However, currently there are no gating values, repeated word hit rate or gradient statistics, and it is impossible to assert which specific tokens benefited.

Partial RoPE: Under the conditions of no XSA, no duplication, and Dropout = 0.1, partial rotation improves by 0.009882 BPB compared to full rotation.

Dropout: Under the conditions of no XSA, no replication, and Full RoPE, a 0.1 residual Dropout improved the best EMA score by 0.053805 BPB compared to 0. Without Dropout at the end of training, the EMA BPB was 2.157944, significantly worse than 1.560301 with Dropout, supporting that regularization helps to prevent the deterioration of validation performance.

5. Number of XSA layers & optimizer comparison

XSA last N layer	Validation BPB	Improvement compared to the absence of XSA
0	1.476084	0.000000
2	1.475453	0.000630
4	1.472912	0.003172
6	1.462978	0.013106
8	1.467045	0.009038

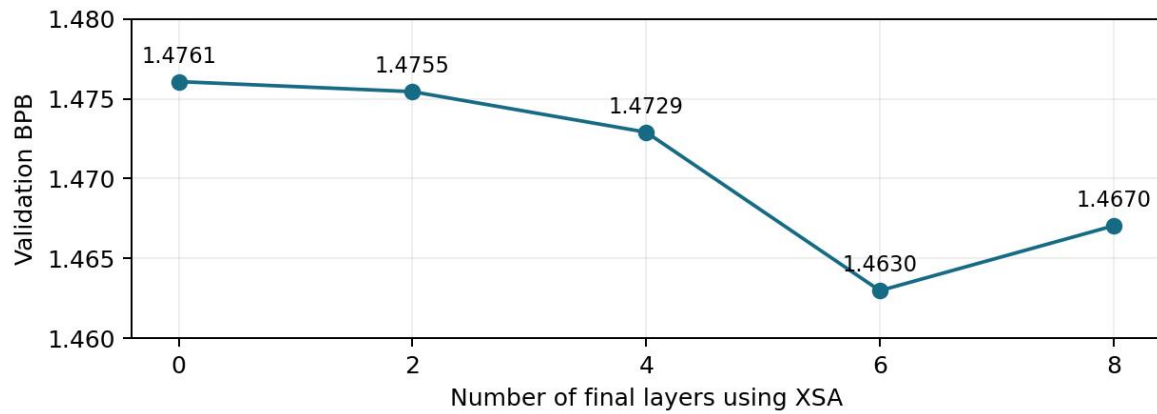


Fig 1 Under the given training budget, the number of XSA layers and the CPU-based validation of BPB at the optimal EMA checkpoint

A 6-layer XSA architecture improves BPB by 0.013106 compared to no XSA at all, and by 0.004068 compared to a full 8-layer configuration. The performance gain does not increase monotonically with the number of coverage layers. One possible explanation is that shallow layers retain standard attention mechanisms, while deeper layers suppress their own value directions, thereby establishing a more appropriate division of labor.

XSA 8 formula	CPU Validation BPB
Muon + AdamW	1.467045
Only AdamW	1.471592

The muon hybrid formulation improved the performance by 0.004546 BPB. Muon demonstrated slightly superior performance compared to the fully tuned AdamW baseline.

6. EMA, training length, and signs of overfitting

Model	Optimal raw BPB / Step	Optimal EMA BPB / Step	Improvement
XSA 6	1.489692 / 1500	1.462978 / 1500	0.026714
XSA 8	1.494441 / 1800	1.467046 / 1400	0.027396
XSA 8 / AdamW	1.496702 / 2100	1.471592 / 1500	0.025110

This table uses the validation records from the training process in metrics.json. There is a difference of approximately 10^{-7} orders of magnitude compared to the independent CPU score. The raw and EMA values come from the same training trajectory and are not from a re-trained "EMA off" experiment. The optimal steps for each may be different. L6 happens to be at the 1500th step, so the two weight versions at the same step can be directly compared.

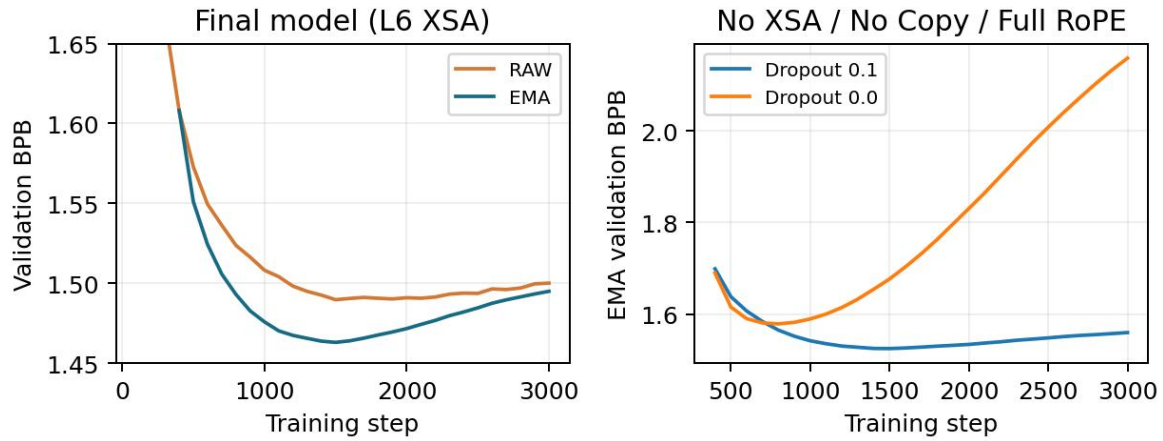


Fig 2 Left: Raw/EMA verification curve of L6 model (vertical axis ranges from 1.45 to 1.65). Right: Dropout control group after adjusting other configurations. The curve is derived from the validation records during the training stage.

The optimal EMA for L6 reached 1.462978 at the 1500th step, then rose to 1.494947 at the 3000th step, with a degradation of 0.031969 BPB. The best point was followed by further updates but did not improve the validation performance, which is consistent with overfitting or deterioration in later generalization. The final score at the last training step cannot be regarded as the final report score.

EMA smooths the update of adjacent parameters, reducing the optimal validation BPB by 0.026714 in this trajectory compared to the raw value. This evidence supports the deployment of EMA weights, but it does not independently examine the initial step and decay coefficient of EMA, and thus cannot confirm that 400 / 0.99 is the global optimum.

7. Computation Costs and Resource Constraints

Experiment	Training target number	Training time / seconds	Total process time / second
XSA all 8	98,304,000	411.78	450.57
MHA + Copy	98,304,000	374.77	412.16
No XSA / Copy	98,304,000	312.43	345.71
Full RoPE	98,304,000	312.28	346.27
No Dropout	98,304,000	308.99	342.47
XSA last 2	98,304,000	385.20	424.88
XSA last 4	98,304,000	394.85	435.35
XSA last 6	98,304,000	402.78	441.62
AdamW	98,304,000	377.47	415.42

Nine groups of formal melting processed a total of 884,736,000 training targets, with cumulative training time of 3280.56 seconds and total process time of 3614.44 seconds. The latter includes data preparation, training, validation during training periods, and checkpoint saving, excluding separately executed CPU scoring.

Resource Item	Record / Calculation	Conclusion Scope
Total CPU test time	27.334s / 5.788s = 4.723×	< 5×
Freeze checkpoint	29,717,047 B = 28.34 MiB	< 64 MiB
Checkpoint, core code, and tokenizer	28.48 MiB	< 64 MiB
CPU peak RAM	2.104GiB	< 4 GiB

The CPU test threshold is set at 5 times the local baseline of 28.939 seconds. The current L6 record stands at 27.334 seconds, leaving little margin. All other indicators are within acceptable limits and have substantial room for tolerance.

8. Reproducibility, Limitations, and Conclusions

Final result files: “code/runs/ablation-L6-xsa-s17/validation_cpu_fp32.json” and “test_cpu_fp32.json”. The “checkpoint_sha256” values of both files match the “best.ema” value in “metrics.json”. The final checkpoint is “best_ema_checkpoint.pt”.

Reproduce:

Reproduce scoring command (execute from code/. The following commands are for reproduction purposes only and were not executed in generating this report):

```
python evaluate.py --checkpoint runs/ablation-L6-xsa-s17/best_ema_checkpoint.pt --split test --output runs/ablation-L6-xsa-s17/reproduced_test.json
```

When re-testing the RAM, add “/usr/bin/time -v” before the above command and save the “Maximum resident set size” in stderr. 4 GiB corresponds to 4,194,304 KiB. This tool provides the peak RSS for the entire evaluation process, and the scoring time still takes the “seconds” of JSON. The baseline should also be scored under the same device, thread and environment.

The current default configuration of student.py is consistent with the configuration of the L6 experiment. To reproduce the training, you can run the command “**python train.py --implementation student --device cuda --precision fp32 --threads 16 --seed 17 --steps 3000 --batch-size 128 --eval-every 100 --run-dir runs/reproduce-L6-s17**”. It is necessary to fix the TRAIN_OPTIONS within the code as well, and not just record the command line.

Result:

The current experiments with the same budget support the use of the combination of the last 6 layers of XSA, replication branches, Dropout, and EMA. The complete test of BPB is 1.477793. There is a trade-off in the number of XSA layers, and replication and regularization yield greater benefits in the corresponding condition comparisons.

Reference

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